



Benchmark Results and Statistics

Exact 43-model common-recipe motor-imagery EEG development benchmark

Exploratory opened-development-cohort evidence only. The observed leader was outcome-selected; this report is not confirmation, clinical validation, proof of novelty, or a global state-of-the-art claim.

Common grid	43 models x 448 participant/fold units x 5 seeds
Exact completion	96,320 / 96,320 jobs; exact audit passed
Primary outcome	Equal-dataset macro balanced accuracy
Plan identity	d4dd852fc9e7b80c8c10c54df03d63ce0ae7e7555ee1b7abdee8d79dca28b61a
Analysis manifest	960ca912d058a2ad3b60def4116dbba436f8ae9162916cde42a4ab265bf01f5d

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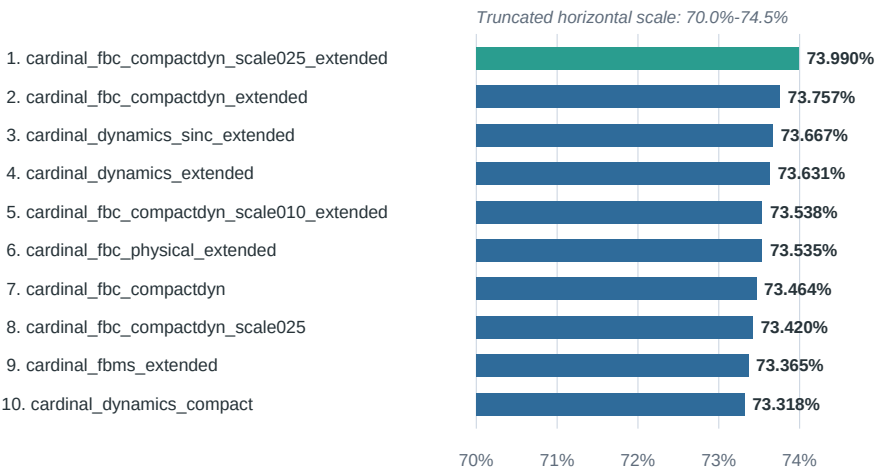
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Executive result

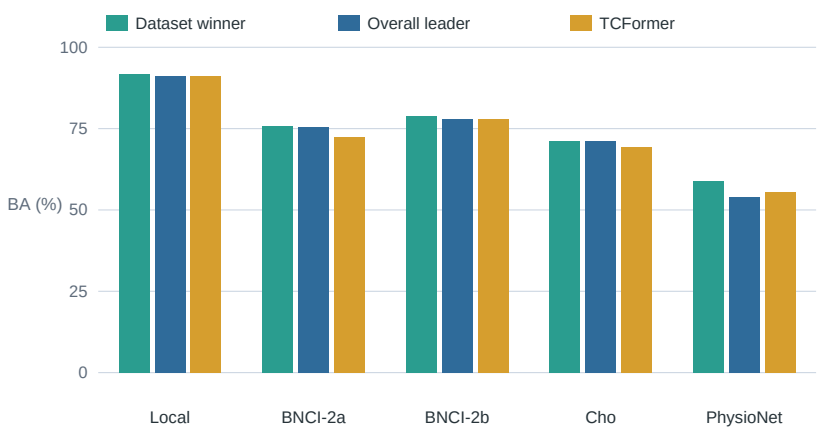
Exact jobs 96,320 / 96,320 zero missing or failed	Observed leader 73.990% cardinal_fbc_compactdyn_scale025_extended	TCFormer 73.316% rank 12 of 43	Leader - TCFormer +0.674 pp 95% descriptive interval [-0.427, +1.899]
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The selected leader's descriptive interval versus TCFormer includes zero. The same leader also had worse aggregate NLL, Brier score, and 15-bin ECE than TCFormer. Rank is not a clinical or statistical separation for every pair.

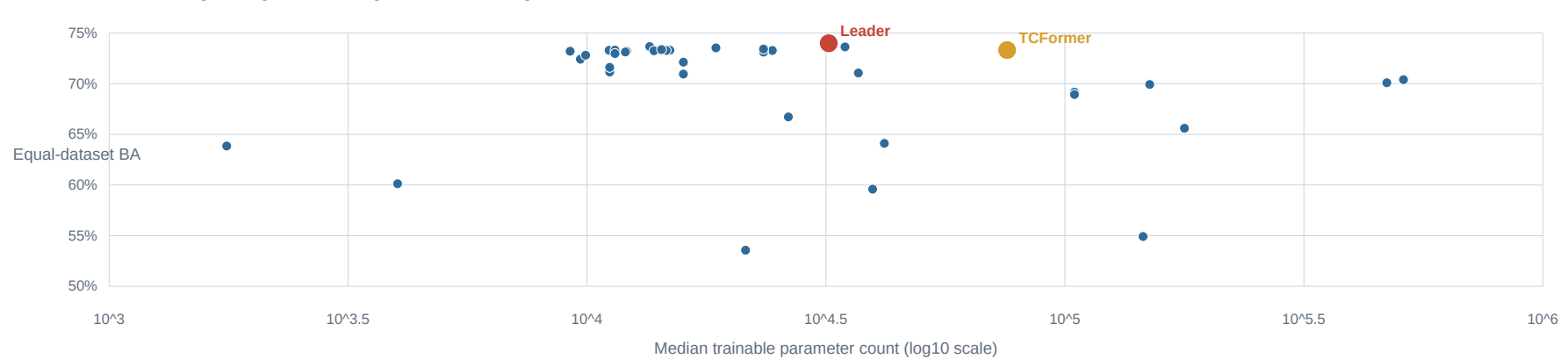
Top 10 equal-dataset balanced accuracy



Dataset behavior: winner, overall leader, TCFormer



Performance-complexity context (all 43 models)



Top 10 common-recipe configurations

Rank	Model	Equal-dataset BA	Evidence status
1	cardinal_fbc_compactdyn_scale025_extended	73.9900%	descriptive opened development
2	cardinal_fbc_compactdyn_extended	73.7571%	descriptive opened development
3	cardinal_dynamics_sinc_extended	73.6674%	descriptive opened development
4	cardinal_dynamics_extended	73.6314%	descriptive opened development
5	cardinal_fbc_compactdyn_scale010_extended	73.5382%	descriptive opened development
6	cardinal_fbc_physical_extended	73.5349%	descriptive opened development
7	cardinal_fbc_compactdyn	73.4642%	descriptive opened development
8	cardinal_fbc_compactdyn_scale025	73.4204%	descriptive opened development
9	cardinal_fbms_extended	73.3652%	descriptive opened development
10	cardinal_dynamics_compact	73.3180%	descriptive opened development

Per-dataset winners

Dataset	Observed winner	Balanced accuracy	Participants
Local	cardinal_dynamics_sinc_extended	91.6250%	8
BNCI-2a	cardinal_fbc_compactdyn_extended	75.7330%	9
BNCI-2b	cardinal_dynamics_sinc_extended	78.9623%	9
Cho	cardinal_fbc_compactdyn_scale025_extended	71.2455%	52
PhysioNet	cardinal_dynamics	58.8414%	54

Descriptive uncertainty versus TCFormer

Rank	Model	Delta (pp)	Fixed-suite descriptive 95% interval (pp)	Resamples
1	cardinal_fbc_compactdyn_scale025_extended	+0.674	[-0.427, +1.899]	100,000
2	cardinal_fbc_compactdyn_extended	+0.441	[-0.619, +1.564]	100,000
3	cardinal_dynamics_sinc_extended	+0.351	[-0.778, +1.518]	100,000
4	cardinal_dynamics_extended	+0.315	[-0.862, +1.582]	100,000
5	cardinal_fbc_compactdyn_scale010_extended	+0.222	[-0.861, +1.391]	100,000
6	cardinal_fbc_physical_extended	+0.219	[-0.833, +1.380]	100,000
7	cardinal_fbc_compactdyn	+0.148	[-0.978, +1.272]	100,000
8	cardinal_fbc_compactdyn_scale025	+0.104	[-0.942, +1.209]	100,000
9	cardinal_fbms_extended	+0.049	[-1.035, +1.231]	100,000
10	cardinal_dynamics_compact	+0.002	[-1.056, +1.047]	100,000

Intervals preserve participant clustering across this fixed opened suite. They are not selection-adjusted confirmatory confidence intervals; no p-values or null-hypothesis decisions were computed.

Calibration and probabilistic quality

Model	BA	Accuracy	NLL (lower)	Brier (lower)	ECE-15 (lower)
cardinal_fbc_compactdyn_scale025_extended	73.99%	74.02%	0.6261	0.3692	14.29%
cardinal_fbc_compactdyn_extended	73.76%	73.77%	0.6641	0.3804	14.79%
cardinal_dynamics_sinc_extended	73.67%	73.67%	0.5149	0.3296	9.08%
fbmsnet	73.32%	73.38%	0.6233	0.3692	13.39%
fbcnnet	73.21%	73.24%	0.6407	0.3777	14.29%
tcformer	73.32%	73.36%	0.5032	0.3212	6.98%

The balanced-accuracy leader is not the calibration leader. Confidence thresholds, abstention, and command activation require prospective calibration in the intended population and operating stream.

Complexity and runtime context

Model	Median parameters	Median inference (ms/trial)	Median peak CUDA (MiB)	Median job time (s)
cardinal_fbc_compactdyn_scale025_extended	32,042	0.1210	236.3	3.66
cardinal_fbc_compactdyn_extended	32,042	0.1202	236.3	2.86
cardinal_dynamics_sinc_extended	13,523	0.0554	134.1	2.19
fbmsnet	11,441	0.1066	318.1	3.64
fbcnnet	9,218	0.0798	198.1	1.92
tcformer	75,668	1.7390	606.6	42.96

Parameter count varies with channel count and class count; the table reports medians across jobs. Shared-workstation load affects timing, so runtime is an engineering outcome, not a scientific tie-breaker.

Full 43-model balanced-accuracy table

Percentages are rounded to two decimals for presentation. The sealed CSV retains full float precision. Overall is an equal average of five dataset means.

Rank	Model	Local	BNCI-2a	BNCI-2b	Cho	PhysioNet	Overall
1	cardinal_fbc_compactdyn_scale025_extended	91.25%	75.46%	77.92%	71.25%	54.07%	73.99%
2	cardinal_fbc_compactdyn_extended	90.21%	75.73%	77.61%	71.15%	54.08%	73.76%
3	cardinal_dynamics_sinc_extended	91.62%	71.02%	78.96%	69.23%	57.50%	73.67%
4	cardinal_dynamics_extended	90.62%	70.61%	78.13%	70.21%	58.57%	73.63%
5	cardinal_fbc_compactdyn_scale010_extended	91.00%	74.81%	77.34%	69.99%	54.55%	73.54%
6	cardinal_fbc_physical_extended	90.79%	74.78%	77.62%	70.11%	54.38%	73.53%
7	cardinal_fbc_compactdyn	88.42%	75.35%	78.19%	71.12%	54.24%	73.46%
8	cardinal_fbc_compactdyn_scale025	89.62%	75.49%	77.10%	70.99%	53.90%	73.42%
9	cardinal_fbms_extended	90.88%	74.39%	77.61%	68.95%	55.01%	73.37%
10	cardinal_dynamics_compact	88.92%	69.84%	78.81%	70.28%	58.74%	73.32%
11	fbmsnet	90.96%	74.22%	77.48%	68.93%	55.00%	73.32%
12	tcformer	91.21%	72.38%	78.00%	69.47%	55.52%	73.32%
13	cardinal_fbc_corr_extended	90.33%	74.80%	75.94%	70.04%	55.40%	73.30%
14	cardinal_fbc_micro	89.08%	75.13%	77.62%	69.96%	54.67%	73.29%
15	cardinal_fbc_micro_extended	90.75%	75.08%	76.22%	70.23%	54.10%	73.28%
16	cardinal_dynamics	88.83%	70.08%	78.20%	70.44%	58.84%	73.28%
17	cardinal_fbc_physical	89.50%	75.13%	77.53%	70.03%	54.04%	73.25%
18	cardinal_fbc_extended	90.67%	74.62%	76.70%	69.66%	54.42%	73.22%
19	fbcnnet	90.62%	74.56%	76.82%	69.60%	54.43%	73.21%
20	cardinal_fbc	89.83%	74.99%	77.18%	69.54%	54.46%	73.20%
21	cardinal_fbc_corr	88.88%	74.89%	76.44%	70.16%	55.28%	73.13%
22	cardinal_fbc_compactdyn_scale010	89.42%	75.11%	76.72%	69.95%	54.37%	73.11%
23	cardinal_fbms	89.21%	74.41%	77.33%	68.94%	54.99%	72.98%
24	cardinal_dynamics_sinc_residual	87.13%	70.64%	78.92%	69.54%	57.84%	72.81%
25	cardinal_dynamics_sinc	87.21%	70.42%	78.80%	68.87%	56.78%	72.42%
26	free_cardinal	88.54%	70.03%	76.04%	67.78%	58.22%	72.12%
27	cardinal_mix_drop	83.17%	72.09%	77.70%	68.10%	57.00%	71.61%
28	cardinal_mix	82.12%	71.98%	77.68%	67.19%	56.77%	71.15%
29	shallow	87.67%	65.44%	77.39%	68.17%	56.61%	71.05%
30	cardinal	83.92%	70.10%	74.62%	67.85%	58.26%	70.95%
31	eegconformer	88.96%	61.81%	78.20%	65.92%	57.09%	70.39%
32	eegconformer_compact	87.38%	62.60%	77.71%	65.47%	57.30%	70.09%
33	ctnet	86.92%	66.00%	78.65%	65.11%	52.93%	69.92%
34	atcnnet	83.92%	62.77%	78.15%	65.24%	55.77%	69.17%
35	atcnnet_aggressive_pool	86.71%	61.92%	77.96%	63.47%	54.59%	68.93%
36	ctnet_compact	82.92%	58.70%	77.77%	61.72%	52.48%	66.72%

Rank	Model	Local	BNCI-2a	BNCI-2b	Cho	PhysioNet	Overall
37	eegsym_wide	86.21%	51.44%	78.30%	60.64%	51.42%	65.60%
38	free_scope	77.42%	55.20%	73.85%	59.79%	54.29%	64.11%
39	eegnet	83.62%	47.53%	72.79%	63.31%	51.98%	63.85%
40	eegtcnet	70.83%	47.73%	73.13%	56.74%	52.19%	60.13%
41	scope	70.79%	43.59%	74.31%	56.95%	52.30%	59.59%
42	deep4	66.62%	31.27%	70.03%	55.48%	51.16%	54.91%
43	eegsym	62.62%	38.32%	63.07%	53.60%	50.20%	53.56%

Full 43-model standard-accuracy table

Accuracy rank is a descriptive secondary ordering. The primary balanced-accuracy rank is retained beside it.

Acc. rank	BA rank	Model	Local	BNCI-2a	BNCI-2b	Cho	PhysioNet	Overall
1	1	cardinal_fbc_compactdyn_scale025_extended	91.25%	75.46%	77.92%	71.25%	54.22%	74.02%
2	2	cardinal_fbc_compactdyn_extended	90.21%	75.73%	77.61%	71.15%	54.12%	73.77%
3	3	cardinal_dynamics_sinc_extended	91.62%	71.02%	78.96%	69.23%	57.53%	73.67%
4	4	cardinal_dynamics_extended	90.62%	70.61%	78.13%	70.21%	58.74%	73.66%
5	5	cardinal_fbc_compactdyn_scale010_extended	91.00%	74.81%	77.34%	69.99%	54.75%	73.58%
6	6	cardinal_fbc_physical_extended	90.79%	74.78%	77.62%	70.11%	54.54%	73.57%
7	7	cardinal_fbc_compactdyn	88.42%	75.35%	78.19%	71.12%	54.31%	73.48%
8	8	cardinal_fbc_compactdyn_scale025	89.62%	75.49%	77.10%	70.99%	54.09%	73.46%
9	9	cardinal_fbms_extended	90.88%	74.39%	77.61%	68.95%	55.31%	73.43%
10	11	fbmsnet	90.96%	74.22%	77.48%	68.93%	55.30%	73.38%
11	12	tcformer	91.21%	72.38%	78.00%	69.47%	55.74%	73.36%
12	13	cardinal_fbc_corr_extended	90.33%	74.80%	75.94%	70.04%	55.59%	73.34%
13	10	cardinal_dynamics_compact	88.92%	69.84%	78.81%	70.28%	58.81%	73.33%
14	14	cardinal_fbc_micro	89.08%	75.13%	77.62%	69.96%	54.81%	73.32%
15	15	cardinal_fbc_micro_extended	90.75%	75.08%	76.22%	70.23%	54.27%	73.31%
16	16	cardinal_dynamics	88.83%	70.08%	78.20%	70.44%	58.95%	73.30%
17	17	cardinal_fbc_physical	89.50%	75.13%	77.53%	70.03%	54.18%	73.28%
18	18	cardinal_fbc_extended	90.67%	74.62%	76.70%	69.66%	54.60%	73.25%
19	19	fbcnnet	90.62%	74.56%	76.82%	69.60%	54.60%	73.24%
20	20	cardinal_fbc	89.83%	74.99%	77.18%	69.54%	54.63%	73.24%
21	21	cardinal_fbc_corr	88.88%	74.89%	76.44%	70.16%	55.47%	73.17%
22	22	cardinal_fbc_compactdyn_scale010	89.42%	75.11%	76.72%	69.95%	54.55%	73.15%
23	23	cardinal_fbms	89.21%	74.41%	77.33%	68.94%	55.29%	73.04%
24	24	cardinal_dynamics_sinc_residual	87.12%	70.64%	78.92%	69.54%	57.93%	72.83%
25	25	cardinal_dynamics_sinc	87.21%	70.42%	78.80%	68.87%	56.94%	72.45%
26	26	free_cardinal	88.54%	70.03%	76.04%	67.78%	58.42%	72.16%
27	27	cardinal_mix_drop	83.17%	72.09%	77.70%	68.10%	57.15%	71.64%
28	28	cardinal_mix	82.12%	71.98%	77.68%	67.19%	56.90%	71.17%
29	29	shallow	87.67%	65.44%	77.39%	68.17%	56.72%	71.08%
30	30	cardinal	83.92%	70.10%	74.62%	67.85%	58.46%	70.99%
31	31	eegconformer	88.96%	61.81%	78.20%	65.92%	57.22%	70.42%
32	32	eegconformer_compact	87.38%	62.60%	77.71%	65.47%	57.53%	70.14%
33	33	ctnet	86.92%	66.00%	78.65%	65.11%	52.99%	69.93%
34	34	atcnet	83.92%	62.77%	78.15%	65.24%	55.92%	69.20%
35	35	atcnet_aggressive_pool	86.71%	61.92%	77.96%	63.47%	54.77%	68.97%
36	36	ctnet_compact	82.92%	58.70%	77.77%	61.72%	52.89%	66.80%

Acc. rank	BA rank	Model	Local	BNCI-2a	BNCI-2b	Cho	PhysioNet	Overall
37	37	eegsym_wide	86.21%	51.44%	78.30%	60.64%	51.74%	65.67%
38	38	free_scope	77.42%	55.20%	73.85%	59.79%	54.40%	64.13%
39	39	eegnet	83.62%	47.53%	72.79%	63.31%	52.42%	63.94%
40	40	eegtcnet	70.83%	47.73%	73.13%	56.74%	52.30%	60.15%
41	41	scope	70.79%	43.59%	74.31%	56.95%	52.34%	59.60%
42	42	deep4	66.62%	31.27%	70.03%	55.48%	51.53%	54.99%
43	43	eegsym	62.62%	38.32%	63.07%	53.60%	50.40%	53.60%

Separate protocols and negative results

Track	Status	Completed / planned	Primary result	Comparison boundary
common raw-trial recipe 43-model grid	complete_exact_audit_pass	96,320 / 96,320	cardinal_fbc_compactdyn_scale025_extended: 73.990%	only exact five-dataset architecture leaderboard
GaugeQuotient opened-development Gate 1	complete_gate_failed	17 / 17	gauge_quotient_crossmoment_v1: 75.070%	separate 17-subject one-seed gate; not comparable to common-grid ranking
CardinalFBMS native-montage transfer	complete_post_outcome_verification_pass	8,675 / 8,675	pretrained_cardinal_fbms: 66.079%	separate target-transfer protocol; post-outcome verification is not the frozen gate
author-recipe adapted TCFormer and FBCNet	pending_unrun	0 / 4,480	-	separate author-recipe reference table
GeoAdapt harmonized-v2 procedures	pending_unrun	0 / 6,495	-	separate procedure table
CAMEO HemiParity PARITY-Fuse and ORBIT-v3 procedures	pending_unrun	0 / 8,780	-	separate binary-task procedure table
deterministic classical controls	pending_unrun	0 / 1,344	-	separate seedless control table
HemiQ-Field harmonized-v2 adaptation	pending_unrun	0 / 45	-	single-dataset procedure table
CardinalFBC native-transfer family	incomplete_not_imported	364 / 6,940	-	separate target-transfer protocol after remaining records
GaugeQuotient disjoint Gate 2	not_run_by_prespecified_rule	0 / not specified	-	Gate 1 failure required candidate termination

Gauge Gate 1 reached 75.0704% on its separate 17-subject, one-seed screen but failed the frozen breadth rule: only two of five dataset deltas were nonnegative. Gate 2 was not run, and Gauge is not a 44th common-grid row.

CardinalFBMS transfer independently verified 8,675 records and a 66.079% equal-dataset condition-mean BA for pretrained CardinalFBMS. It is a separate two-dataset native-transfer protocol; the post-outcome verifier is explicitly not the frozen gate.

Interpretation caveats

- All five common cohorts were previously opened development cohorts.
- The winner was outcome-selected from 43 configurations; its interval is descriptive, not confirmatory.
- The five dataset environments do not represent the global population of montages, users, or care settings.
- Most recordings are from healthy volunteers; no patient-facing efficacy or safety claim is supported.
- Pseudonymous job-, subject-seed-, and subject-level metrics remain participant-derived and require governance review before public release.
- Raw EEG, trial predictions, direct identifiers, embeddings, and checkpoints are not included.
- Extended/compact/scale variants are configurations or ablations, not automatically separate novel architectures.
- Calibration, false activation, abstention, fatigue, latency, drift, and failure modes require prospective evaluation.

Provenance

Artifact	SHA-256 or identity
Plan identity	d4dd852fc9e7b80c8c10c54df03d63ce0ae7e7555ee1b7abdee8d79dca28b61a
Preflight report	39bef71293d33f343eb2d908f0c9bdc6cef06c9e457b98575ca06e86c80af4ef
Final audit	010b44542157efb6ba5a95376198b9b691e9803ce3fe605e77cb5cdb9739d063
Analysis manifest	960ca912d058a2ad3b60def4116dbba436f8ae9162916cde42a4ab265bf01f5d
Analysis contract	6eb975790bf7213e614159172d673b5e18ebda554f674e01c5b159099273393e
Analysis input ledger	3f68621a316a6a90cf18ce073e05f70b453a13fa7ef4407f66a90f54562dcbbf
Forensic ledger	962531ea433232d90d9869a2d78529f2e867cd7373b883dd78e8cf01ca31f626